

CO1_ASSESSMENT_2

Question 1: Mapping Data onto Aesthetics

Program:

```
library(ggplot2)

data <- data.frame(

  Product = c("P1","P2","P3","P4","P5","P6"),

  Sales = c(120,90,150,80,135,100),

  Profit = c(18,12,25,8,22,16),

  Category = c("Electronics","Clothing","Electronics","Furniture","Furniture","Clothing")

)

ggplot(data, aes(x = Sales, y = Profit, color = Category, size = Sales)) +

  geom_point(alpha = 0.8) +

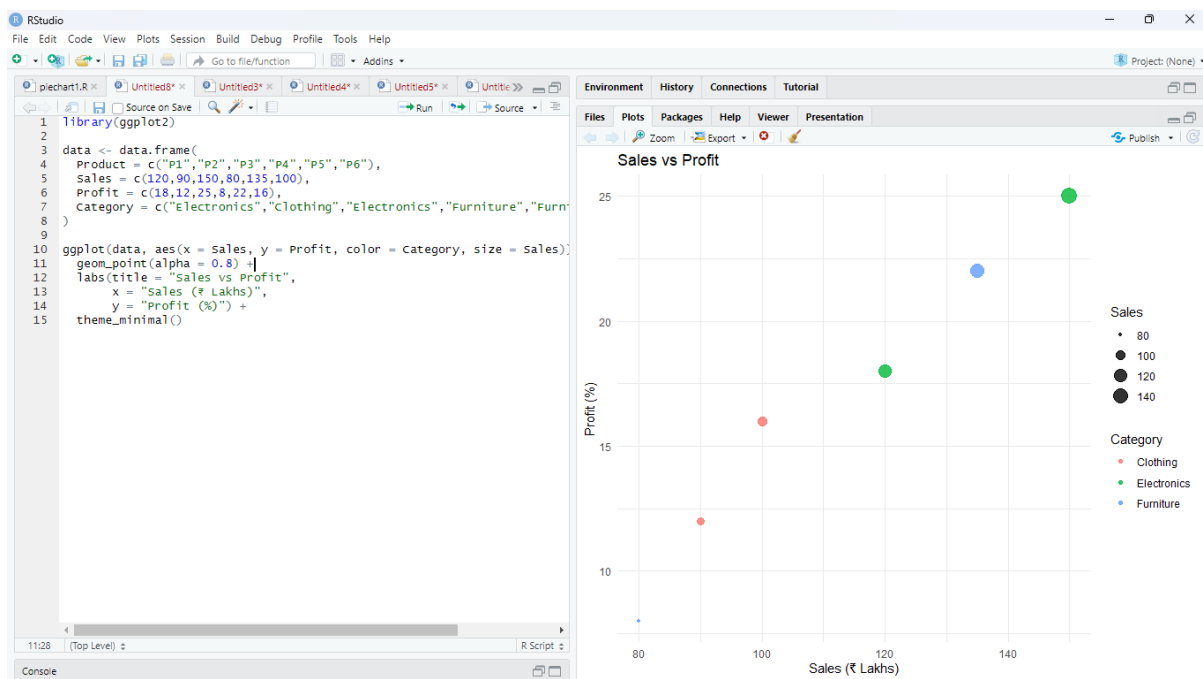
  labs(title = "Sales vs Profit",

       x = "Sales (₹ Lakhs)",

       y = "Profit (%)") +

  theme_minimal()
```

Output:



Question 2: Coordinate Systems (Cartesian and Polar)

Program:

```
library(ggplot2)

data <- data.frame(

  Direction = c("N","NE","E","SE","S","SW","W","NW"),

  Frequency = c(18,12,20,15,22,11,16,14)

)

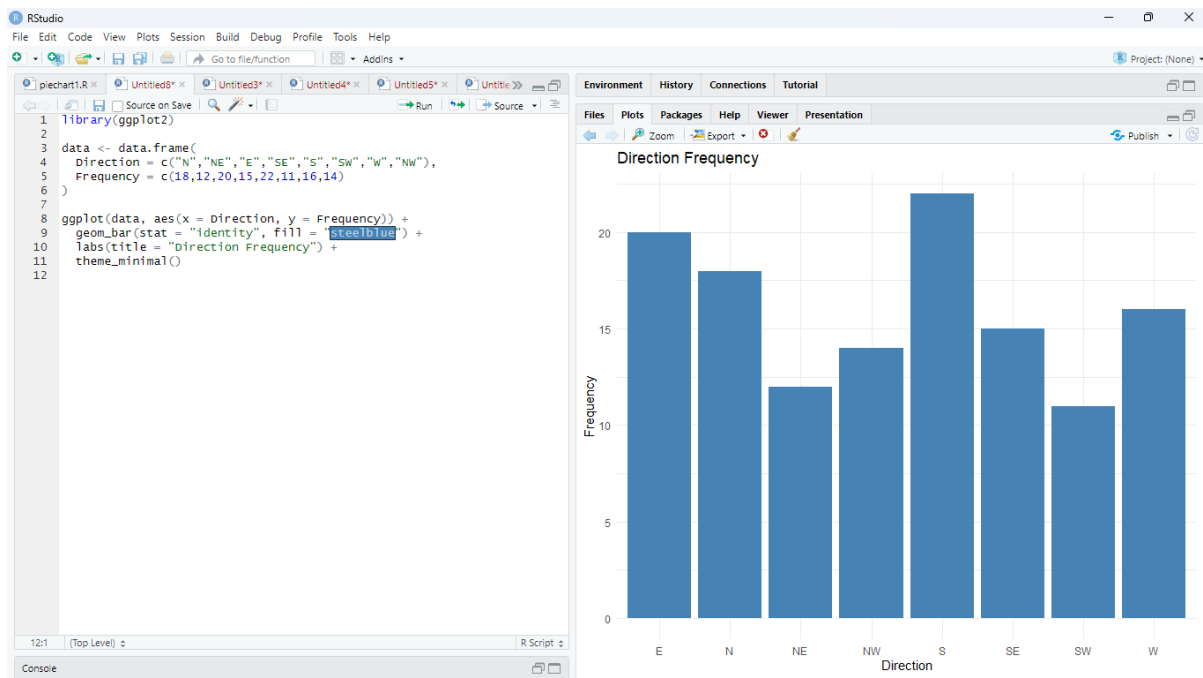
ggplot(data, aes(x = Direction, y = Frequency)) +

  geom_bar(stat = "identity", fill = "steelblue") +

  labs(title = "Direction Frequency") +

  theme_minimal()
```

Output:



Question 3: Color Scales for Data Representation

Program:

```
library(ggplot2)

data <- data.frame(

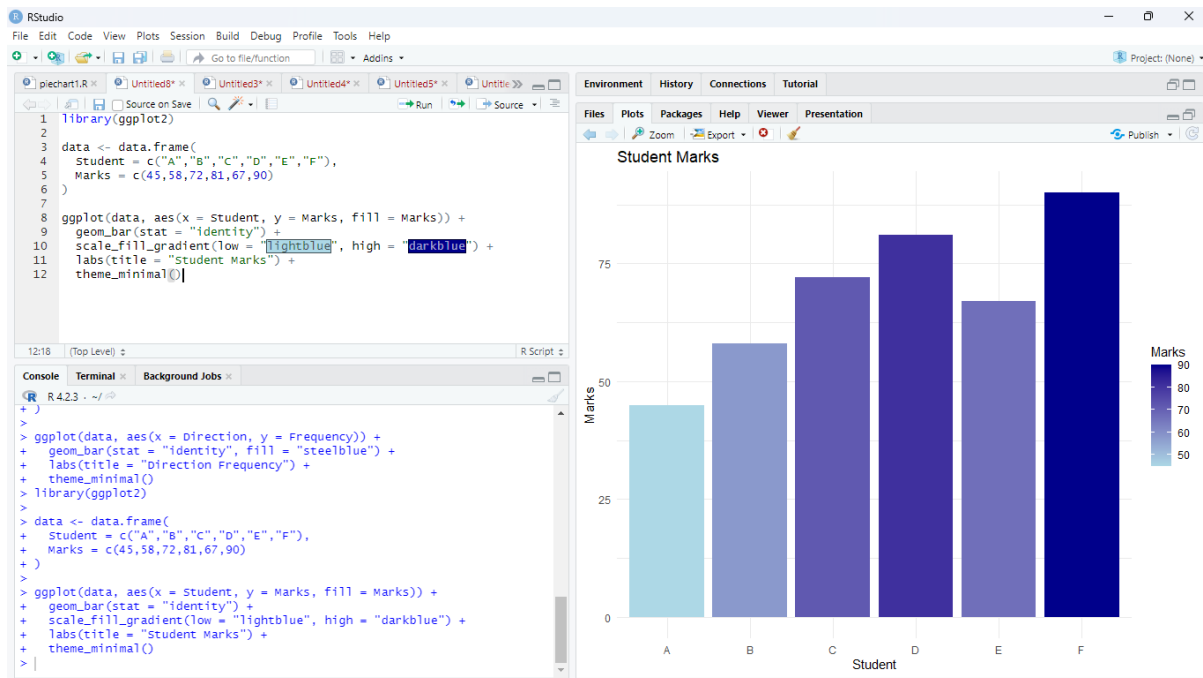
  Student = c("A","B","C","D","E","F"),

  Marks = c(45,58,72,81,67,90)

)
```

```
ggplot(data, aes(x = Student, y = Marks, fill = Marks)) +
  geom_bar(stat = "identity") +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(title = "Student Marks") +
  theme_minimal()
```

Output:



Question 4: Highlighting with Color

Program:

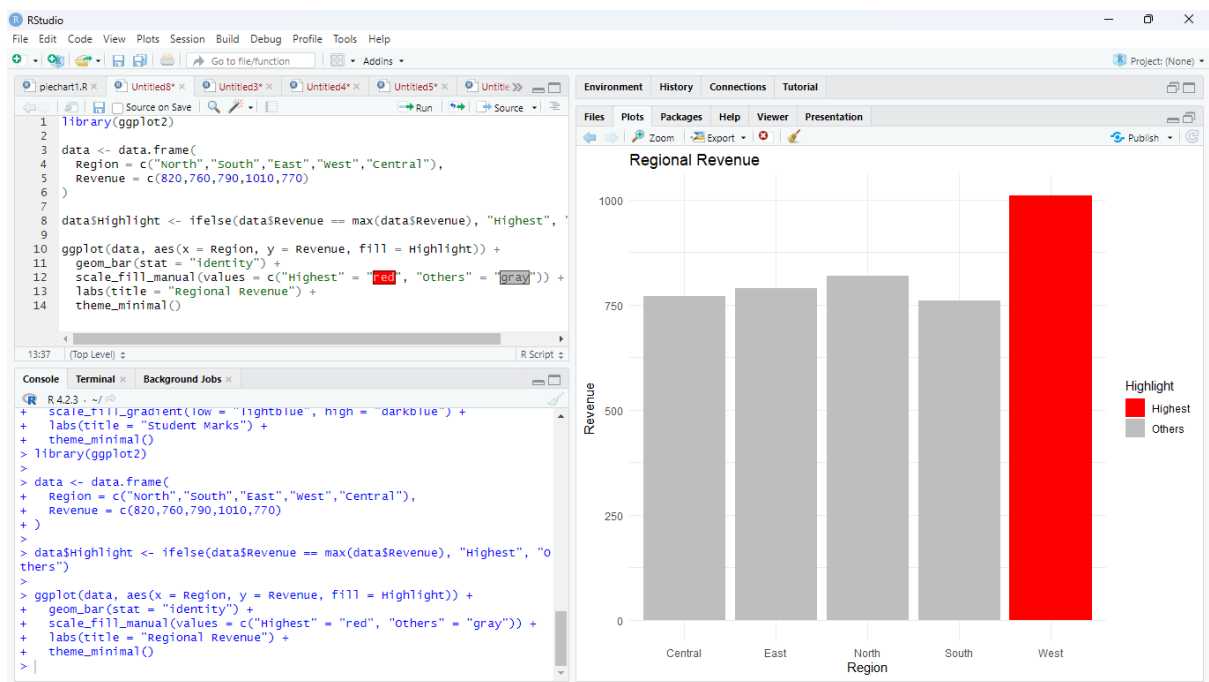
```
library(ggplot2)

data <- data.frame(
  Region = c("North","South","East","West","Central"),
  Revenue = c(820,760,790,1010,770)
)

data$Highlight <- ifelse(data$Revenue == max(data$Revenue), "Highest", "Others")

ggplot(data, aes(x = Region, y = Revenue, fill = Highlight)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("Highest" = "red", "Others" = "gray")) +
  labs(title = "Regional Revenue") +
  theme_minimal()
```

Output:



Question 5: Integrated Visualization

Program:

```
library(ggplot2)

data <- data.frame(
  City = c("A", "B", "C", "D", "E"),
  Population = c(8.2, 5.6, 9.4, 6.8, 4.9),
  AQI = c(110, 180, 90, 240, 150),
  Rainfall = c(850, 430, 1200, 520, 680),
  Region = c("North", "South", "North", "East", "West")
)

ggplot(data,
  aes(x = Population,
    y = AQI,
    size = Rainfall,
    color = Region)) +
  geom_point(alpha = 0.8) +
  labs(title = "Population vs AQI",
    x = "Population (Million)",
```

```
y = "AQI") +  
theme_minimal()
```

Output:

